

Dynatrace Workflows

Series: AUTOM | Notebook: 5 of 8 | Created: January 2026

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Prerequisites

Before starting this notebook, ensure you have:

Requirement	Description
Dynatrace SaaS	Tenant with Workflows enabled
Permissions	AutomationWorkflows permission
Basic DQL	Understanding of Dynatrace Query Language

Learning Objectives

By the end of this notebook, you will:

- Understand Dynatrace Workflows architecture
 - Know how to create event-driven automations
 - Be able to implement auto-remediation patterns
 - Connect workflows to external systems
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1. Introduction

Dynatrace Workflows is a built-in automation engine that enables event-driven actions directly within the platform. Unlike external tools, workflows run inside Dynatrace with full access to observability data.

Why Workflows?

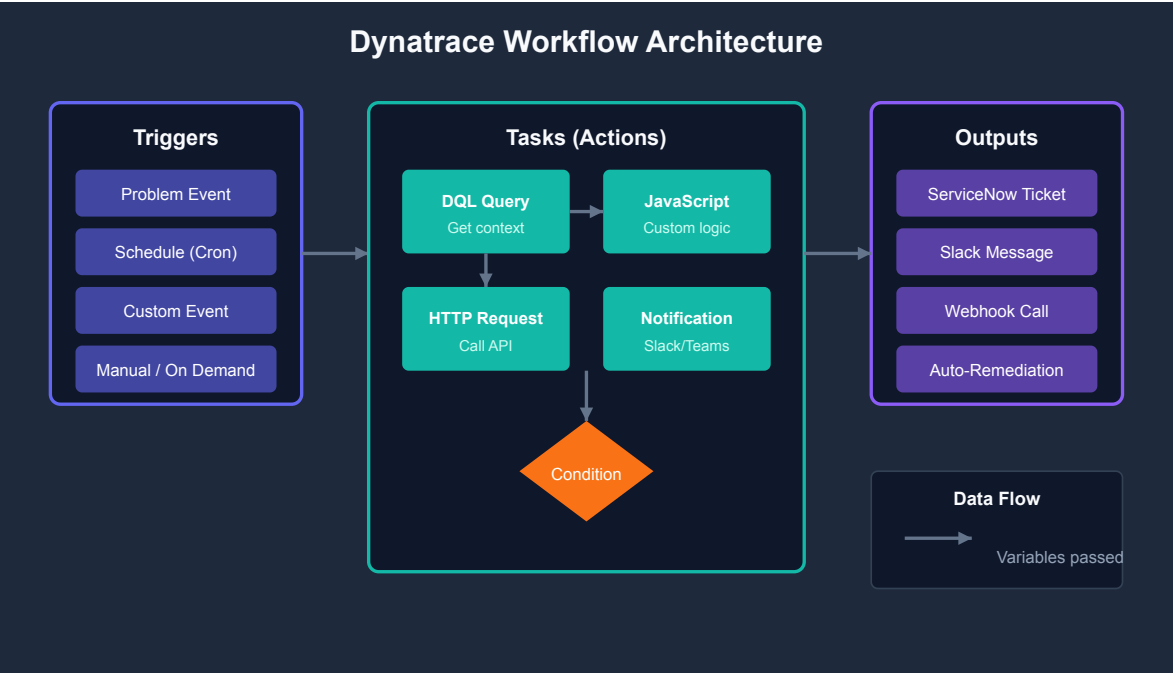
Benefit	Description
Native Integration	Direct access to all Dynatrace data
No Infrastructure	No external systems to manage
Event-Driven	React to problems, alerts, schedules
Context-Aware	Full Davis AI context available
Secure	Credentials stored in Dynatrace vault

Common Use Cases

Use Case	Description
Notifications	Send alerts to Slack, Teams, email
Auto-Remediation	Restart services, scale resources
Ticketing	Create ServiceNow, Jira tickets
Enrichment	Add context to problems
Reporting	Generate scheduled reports

2. Workflow Components

Workflow Architecture



Triggers

Trigger Type	Description	Example
Problem	Davis problem event	Availability issue detected
Event	Custom or ingest event	Deployment completed
Schedule	Time-based (cron)	Daily at 9 AM
Manual	User-initiated	On-demand execution
On Demand	API or SDK call	External system trigger

Tasks

Task Type	Description
HTTP Request	Call external APIs
Run JavaScript	Custom logic
DQL Query	Query Grail data
Send Notification	Slack, Teams, email
Create Issue	Jira, ServiceNow
Run Workflow	Call other workflows

3. Creating Workflows

Access Workflows

Navigate to: **Apps → Workflows**

Or via URL:

`https://{tenant}.apps.dynatrace.com/platform/app/dynatrace.automations/workflows`

Simple Notification Workflow

Workflow YAML (exportable):

```
title: Problem Notification
description: Send Slack notification for critical problems
trigger:
  type: davis-problem
  config:
    categories:
      - AVAILABILITY
      - ERROR
tasks:
  send_slack:
    name: Send Slack Alert
    action: dynatrace.slack:send-message
    input:
      connection: slack-webhook
      channel: "#alerts"
      message: |
        :warning: *Problem Detected*
        *Title:* {{ event().title }}
        *Severity:* {{ event().severity }}
        *Impact:* {{ event().impactLevel }}
```

Problem Trigger Configuration

Option	Description
Categories	AVAILABILITY, ERROR, SLOWDOWN, RESOURCE, CUSTOM
Severities	Filter by severity level
Management Zones	Scope to specific zones
Entity Types	Filter by entity (HOST, SERVICE, etc.)

Using JavaScript Tasks

Custom logic with JavaScript:

```
// Task: Enrich problem data
export default async function ({ execution_id }) {
  // Access previous task output
  const problem = execution.result.event;

  // Build enrichment data
  const enrichment = {
    problemId: problem.id,
    title: problem.title,
    affectedEntities: problem.affectedEntities.length,
    rootCauseEntity: problem.rootCauseEntity?.name || 'Unknown',
    timestamp: new Date().toISOString()
  };

  return enrichment;
}
```

Using DQL Tasks

Query data within workflows:

```
tasks:
  get_metrics:
    name: Get CPU Metrics
    action: dynatrace.automations:run-dql-query
    input:
      query: |
        timeseries avg(dt.host.cpu.usage), by:{dt.entity.host}
        | filter dt.entity.host == "{{ event().rootCauseEntity.id }}"
        | limit 10
```

4. Actions and Integrations

Built-in Connectors

Connector	Actions
Slack	Send message, post to channel
Microsoft Teams	Send adaptive card
Jira	Create/update issue
ServiceNow	Create incident, update CI
PagerDuty	Create incident
OpsGenie	Create alert
Email	Send email notification

HTTP Request Task

Call any REST API:

```
tasks:
  call_api:
    name: Call External API
    action: dynatrace.automations:http-function
    input:
      url: "https://api.example.com/incidents"
      method: POST
      headers:
        Content-Type: application/json
        Authorization: "Bearer {{ env.API_TOKEN }}"
      body: |
        {
          "title": "{{ event().title }}",
          "severity": "{{ event().severity }}",
          "source": "dynatrace"
        }
```

Credential Management

Store credentials securely:

1. Go to **Settings → Credential Vault**
 2. Create a new credential
 3. Reference in workflow: `{{ credential.MY_API_KEY }}`
-

5. Auto-Remediation Patterns

Pattern: Restart Service

```
title: Auto-Restart on Crash
trigger:
  type: davis-problem
  config:
    categories:
      - AVAILABILITY
tasks:
  check_entity:
    name: Validate Entity Type
    action: dynatrace.automations:run-javascript
    input:
      script: |
        const entity = execution.result.event.rootCauseEntity;
        if (entity.type !== 'PROCESS_GROUP_INSTANCE') {
          return { skip: true };
        }
        return { entityId: entity.id, skip: false };

  restart_service:
    name: Execute Restart
    action: dynatrace.automations:http-function
    conditions:
      states:
        check_entity: OK
        custom: "{{ result('check_entity').skip == false }}"
    input:
      url: "https://runbook.example.com/restart"
      method: POST
      body: |
        {
          "entityId": "{{ result('check_entity').entityId }}"
        }
```

Pattern: Scale on High CPU

```
title: Auto-Scale on Resource Pressure
trigger:
  type: davis-problem
  config:
    categories:
      - RESOURCE
tasks:
  check_cpu:
    name: Query Current CPU
    action: dynatrace.automations:run-dql-query
    input:
      query: |
        timeseries avg(dt.host.cpu.usage)
        | filter dt.entity.host == "{{ event().rootCauseEntity.id }}"

  scale_up:
    name: Trigger Scale Event
    action: dynatrace.automations:http-function
    conditions:
      custom: "{{ result('check_cpu')[0].avg > 90 }}"
    input:
      url: "https://api.cloud.example.com/scale"
      method: POST
```

Pattern: Create Incident Ticket

```
title: ServiceNow Incident Creation
trigger:
  type: davis-problem
  config:
    categories:
      - AVAILABILITY
      - ERROR
tasks:
  create_incident:
    name: Create ServiceNow Incident
    action: dynatrace.servicenow:create-incident
    input:
      connection: servicenow-prod
      short_description: "Dynatrace: {{ event().title }}"
      description: |
        Problem detected by Dynatrace Davis AI

        Title: {{ event().title }}
        Severity: {{ event().severity }}
        Impact: {{ event().impactLevel }}
        Root Cause: {{ event().rootCauseEntity.name }}

        Link: {{ event().url }}
      urgency: 2
      impact: 2

  add_comment:
    name: Comment on Problem
    action: dynatrace.automations:run-javascript
    input:
      script: |
        const ticketId = execution.result.create_incident.sys_id;
        // Add comment to Dynatrace problem with ticket reference
        return { ticketId, status: 'created' };
```

6. Best Practices

Workflow Design

Practice	Description
Single responsibility	One workflow, one purpose
Idempotency	Safe to run multiple times
Error handling	Handle failures gracefully
Logging	Add descriptive task names
Testing	Use manual triggers for testing

Trigger Configuration

Practice	Description
Scope narrowly	Use management zones and filters
Avoid duplicates	Check if action already taken
Debounce	Add delays for flapping alerts

Security

Practice	Description
Credential vault	Never hardcode secrets
Least privilege	Minimal API permissions
Audit trail	Log all remediation actions

Error Handling

```
tasks:
  api_call:
    name: Call API
    action: dynatrace.automations:http-function
    retry:
      count: 3
      delay: 30s
    on_error:
      next: error_handler

  error_handler:
    name: Handle Error
    action: dynatrace.slack:send-message
    input:
      channel: "#alerts"
      message: "Workflow failed: {{ error().message }}"
```

7. Next Steps

Workflows vs External Automation

Scenario	Tool Choice
Event-driven from Dynatrace	Workflows
Config deployment	Monaco/Terraform
Complex orchestration	External (Ansible, etc.)
Cross-platform actions	Workflows with HTTP tasks

Continue the Series

Next Notebook	Focus
AUTOM-06: Dynatrace SDKs	Programmatic access with TypeScript/Python

Additional Resources

- [Workflows Documentation](#)
 - [Workflow Actions Reference](#)
 - [JavaScript Task Reference](#)
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Summary

In this notebook, you learned:

- Dynatrace Workflows architecture and components
- Creating event-driven automations with triggers and tasks
- Implementing auto-remediation patterns
- Best practices for workflow design and security

Key Takeaway: Workflows provide native automation inside Dynatrace. Use them for event-driven actions like notifications, ticketing, and auto-remediation. For configuration management, continue using Monaco or Terraform.

Continue to AUTOM-06: Dynatrace SDKs to learn programmatic access patterns.

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